

Education

University of Utah
B.S. Computer Science

Salt Lake City, UT
August 2020 – May 2025

Relevant Coursework: Computer Systems, Machine Learning, Computer Graphics, Algorithms, Software Practice I & II, Database Systems, Computer Networks, Foundations of Data Analysis, Models of Computation, and Linear Algebra.

Skills

Languages: Python, TypeScript, C++, SQL, Java

Software: Pandas, NumPy, React, Node.js, AWS, Azure, Docker, Next.js, .NET, REST, Django, MongoDB, Linux

Experience

L3Harris Technologies
Associate Software Engineer

Dallas, TX
June 2025 – Present

- Developed and maintained real-time signal processing software in C++ on embedded Linux platforms, working directly alongside electrical and hardware engineers to integrate software with custom hardware designs and meet strict timing and resource constraints.
- Modified and extended an established C++ codebase for defense sensor systems, exercising judgment on when to modernize versus preserve existing behavior, and exposing clean Python-accessible APIs for test and automation workflows.
- Debugged production issues in closed embedded environments using structured logging and systematic analysis, collaborating in a flat, cross-functional team to bring up new hardware and resolve integration failures without interactive debugging tools.

Doxy.me
Software Engineer

Charleston, SC
August 2024 – May 2025

- Built a real-time mental health inference system for Doxy.me's telehealth platform, fine-tuning Llama models to predict depression risk, anxiety, and mood indicators from tone, voice, and facial cues captured over live WebRTC sessions, served via SageMaker and Fargate.
- Improved prediction accuracy of the mental health inference system by applying RL feedback loops over structured clinical signals and physician-validated outcomes from live video consultations.
- Eliminated environment drift and enabled zero-downtime releases by standardizing CI/CD configuration with YAML-based pipelines across the ML stack.

University of Utah
Undergraduate Research Assistant – FuTURES Lab

Salt Lake City, UT
May 2024 – Jun 2025

- Analyzed and optimized large-scale datasets, enhancing software configuration testing with gcov and CMake, increasing code coverage by 30% on real-world APIs (Libpng, PyTorch).
 - Expanded OSS-Fuzz testing coverage using compile-time options to improve the robustness of full-stack libraries.
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Projects

Configuration Fuzzer: Developed a configuration fuzzing tool for OSS-Fuzz, automating build generation to identify critical compile-time configurations and improve code coverage, uncovering previously untested code paths and increasing bug detection effectiveness.

University Course Planning/Review Platform (*Class Best*): Built a multi-service data platform enabling student schedule matching, course overlap detection, and behavioral insights. Developed ETL pipelines in Python to process thousands of course records, user preferences, and time-series activity logs. Built a responsive frontend in React + Tailwind with interactive dashboards, heatmaps, and calendar visualizations.

Telehealth Mental Health Diagnosis Tool (Doxy.me): Built an AI system that analyzes visual, speech, and tonal cues from telehealth sessions using PyTorch and Llama, delivering structured mental health insights to treating physicians via SageMaker and Fargate inference pipelines. Improved diagnostic accuracy using reinforcement learning and WebRTC video integration.